

Learning accurate neighborhood- and self-information for higher-order relation prediction in Heterogeneous Information Networks

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ARTICLE INFO

Communicated by G. Gnecco

Keywords:

Heterogeneous information networks
Higher-order relation prediction
Graph neural network

ABSTRACT

Heterogeneous Information Networks (HINs) are commonly employed to model complex real-world scenarios with diverse node and edge types. However, due to constraints in data collection and processing, constructed networks often lack certain relations. Consequently, various methods have emerged, particularly recently, leveraging heterogeneous graph neural networks (HGNNs) to predict missing relations. Nevertheless, these methods primarily focus on pairwise relations between two nodes. Real-world interactions, however, often involve multiple nodes and diverse types, extending beyond simple pairwise relations. For instance, academic collaboration networks may entail interactions among authors, papers, and conferences simultaneously. Despite their prevalence, higher-order relations are often overlooked. While HGNNs are effective at learning network structures, they may suffer from over-smoothing, resulting in similar representations for nodes and their neighbors. The learned inaccurate proximity among nodes impedes the discernment of higher-order relations. Furthermore, observed edges among a target group of nodes can provide valuable evidence for predicting higher-order relations. To address these challenges, we propose a novel model called Accurate Neighborhood- and Self-information Enhanced Heterogeneous Graph Neural Network (ANSE-HGN). Building upon HGNNs to encode network structure and attributes, we introduce a relation-based neighborhood encoder to capture information within multi-hop neighborhoods in heterogeneous higher-order relations. This enables the calculation of accurate proximity among target groups of nodes, thereby enhancing prediction accuracy. Additionally, we leverage self-information from observed higher-order relations as an auxiliary loss to reinforce the learning process. Extensive experiments on four real-world datasets demonstrate the superiority of our proposed method in higher-order relation prediction tasks.

1. Introduction

In the real world, networks are widely used to model various scenarios, where entities serve as nodes and interactions are denoted by edges. Heterogeneous Information Networks (HINs), as opposed to homogeneous ones, are distinguished by their diverse types of nodes and edges, making them more capable of modeling complex scenarios such as academic collaboration [1,2], E-commerce recommendation [3,4], and social networks [5,6]. Constraints in data collection and processing, including privacy and security concerns, often lead to missing relations in constructed networks. Nonetheless, leveraging available information enables us to discern laws in relation formation from the network's effective topology, thereby facilitating inference of potential missing relations.

In recent years, many relation prediction methods [7–9] have emerged, focusing mainly on studying the relations between pairs of

nodes. However, real-world interactions often occur within groups, involving multiple nodes and complex patterns. For example, social interactions typically occur within groups [10], biological interactions involve sets of molecules [11], and chemical reactions involve various substances [12]. Intuitively, in the academic collaboration network depicted in Fig. 1(a), when author B publishes paper A at conference C, they form a higher-order relation {A, B, C}. In a word, higher-order relations, involving more than two nodes, are prevalent.

Several studies [13–15] have already revealed the significance of higher-order relations in networks, with some [16,17] further emphasizing that higher-order relations form the backbone of many graph mining problems. Networks often evolve through higher-order relations [14], such as the mentioned academic collaboration networks evolving through paper publications (Fig. 1(a)). When a new paper

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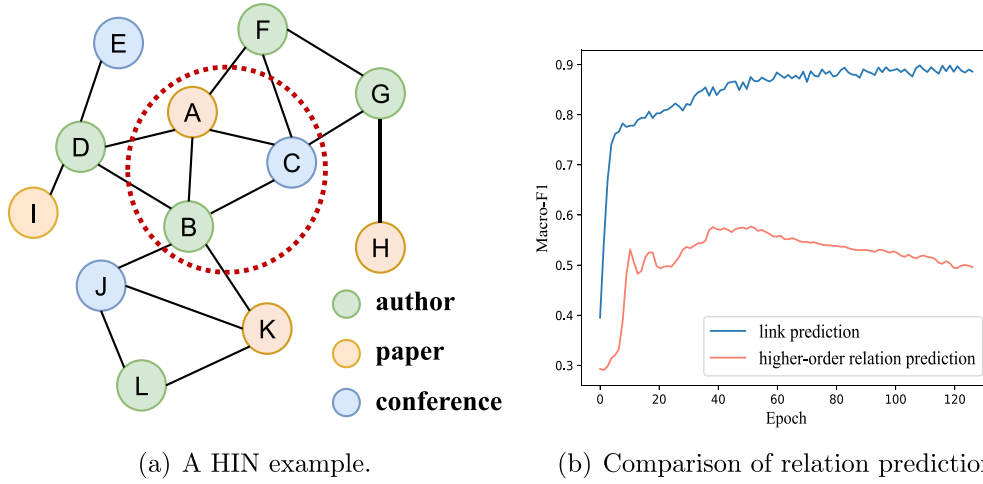


Fig. 1. (a) A HIN example of an academic collaboration network. Different colors represent different node types. The red dashed circle encloses the heterogeneous higher-order relation {A, B, C}. (b) The comparison of performance-changing trends between link prediction and higher-order relation prediction using the same HGNN.

is published, higher-order relations composed of {paper, author, conference} are added to the network as localized structures. Compared to traditional relation prediction, higher-order relation prediction can better reflect the complex interactions between groups in networks, providing a more profound insight into network structure. This enables higher-order relation prediction to support more comprehensive practical applications. For instance, in an academic recommendation system, recommendations can encompass papers, authors, and fields simultaneously, thereby reflecting different aspects of the user's interests; in knowledge graph completion, higher-order relation prediction considers more diverse associations between entities and better utilizes contextual information to improve the accuracy of knowledge graph completion. In summary, higher-order relations extend and generalize traditional pairwise relations, facilitating a more comprehensive understanding of network structure.

The existing methods for higher-order relations prediction can generally be categorized into subgraph prediction [18–20] and hyperedge prediction [10,21,22]. In subgraph prediction, higher-order relations are described using subgraphs such as triangles. For example, literature [18] employs various triangle embedding methods to address higher-order relation prediction problems and SEAM [19] extracts closed subgraphs around nodes of interest to model higher-order relations. To achieve a more granular differentiation of higher-order relations that subgraphs cannot distinguish, some methods utilize hyperedges to describe higher-order relations. For instance, CMM [21] predicts hyperlinks in the adjacency space using the expectation-maximization algorithm. HIT [10] defines a series of refined higher-order patterns with hyperedges and predicts them by encoding context structures. However, these methods are designed for homogeneous networks and cannot be directly applied to HINs. Despite its importance, research on higher-order relation prediction on HINs remains limited.

Heterogeneous Graph Neural Networks (HGNNs) are a classic class of methods for structure learning and relation prediction, using message-passing mechanisms to understand the context of nodes. A plain idea is to directly extend the HGNN learned in link prediction tasks to higher-order relation prediction. We have empirically evaluated this approach. Specifically, we employed a relation-based HGNN trained for link prediction on the DouBan dataset. Then we used it to generate features for higher-order relation prediction. As shown in Fig. 1(b), the performance of link prediction is steadily improving as the epoch increases, demonstrating that HGNNs can capture the proximity between two nodes. However, the performance of higher-order relation prediction first increases and then decreases. This indicates that while pairwise relation prediction serves as inspiration for higher-order relation prediction, the over-smoothing issue of HGNNs causes node representations to become

overly similar to their neighbors, which is insufficient to capture the proximity within higher-order relations. Consequently, the learned inaccurate proximity among nodes fails to effectively infer the relation within a group of nodes from a higher-order level.

To address the issue mentioned above, this paper proposes a novel Accurate Neighborhood- and Self-information Enhanced Heterogeneous Graph Neural Network model for predicting higher-order relations in HINs, named ANSE-HGN. Utilizing a relation-based HGNN, we first encode the structures and attributes of the network with an attention fusion module. Note that various HGNNs can be selected, as our ANSE-HGN is a potent extension of general relation-based HGNNs. To capture the accurate proximity within higher-order relations, we design a structure feature encoder that utilizes multi-hop neighborhood information. This encoder comprises two components. Specifically, a relation-based structure encoder learns features for all types of edges and then aggregates edge information into nodes to derive their structural feature values. This component can autonomously learn and generate beneficial structural feature values without manual design. Then a neighbor-based structure encoder fuses structural features from multi-hop neighbors for each node, thereby generating structure feature vectors enriched with abundant neighborhood information. We jointly calculate the similarity of node pairs in the higher-order relations and incorporate this discrimination information into the base embeddings of HGNN to assist in higher-order relation prediction. In the loss function, we utilize the self-information (connectivity and centrality) of higher-order relations to provide additional assistance in the learning process. Our contributions are as follows:

- (1) We introduce the task of higher-order relation prediction in HINs and present a novel model to address it, at an early stage in this area.
- (2) The proposed model ANSE-HGN effectively captures the proximity within heterogeneous higher-order relations by leveraging multi-hop neighborhood information, overcoming the limitations of HGNNs. Moreover, the new self-information loss further boosts the model's performance by leveraging connectivity and centrality.
- (3) We evaluated our model on four real-world datasets, surpassing state-of-the-art methods, and demonstrating its superiority in predicting higher-order relations on HINs.

The remainder of the paper is organized as follows. In Section 2, we discuss related work on higher-order relation prediction. Section 3 provides an overview of the notations and problem definitions used in the paper. In Section 4, we describe the proposed method in detail. In Section 5, we interpret and analyze the experimental results. Finally, in Section 6, we conclude the paper and discuss its limitations.

2. Related work

In this section, we primarily review research related to higher-order relations, focusing on studies closely connected to our work, including higher-order relation prediction and heterogeneous network representation learning methods. Furthermore, in Section 2.3, we compare the differences between previous research and this paper. The following subsection will provide detailed introductions.

2.1. Higher-order relation prediction

Higher-order relations are ubiquitous, and numerous studies [13–17] have demonstrated their importance in network research. Existing methods for higher-order relation prediction can generally be categorized into subgraph prediction [18–20] and hyperedge prediction [10, 21–24].

In subgraph prediction, higher-order relations are often modeled as triangles, induced subgraphs, and similar structures. For example, literature [18] aims to solve the problem of the closure of open triangles, it encodes triangles and model interactions by extending node2vec, graph2vec, and graph neural networks. SEAM [19] models higher-order relations as motifs. By appropriately selecting negative samples, it captures the richness of various motifs and their surrounding contexts. To provide additional neighborhood information, SEAM introduces two node labeling schemes, which are then jointly fed into a DGCNN [25] for prediction. TRPR [20] is an iterative method, similar to PageRank, designed to predict nodes that can form triangles. It considers edges that participate in multiple triangles as more important for prediction, assigning more weight to these edges and thereby reinforcing the significance of triangle-related edges. Because subgraphs cannot capture certain subtle structures, hyperedges have been introduced to overcome this limitation. In hyperedge prediction, HIT [10] utilizes hyperedges to define four different types of higher-order relations on temporal networks, encoding these relations by capturing contextual proximity to infer real patterns and predict when these interactions might occur. Literature [23] predicts potential new hyperedges in a k -uniform hypergraph by calculating the eigenvectors of the hypergraph Laplacian tensor. These eigenvectors are used to evaluate the construction cost, allowing the selection of hyperedges most likely to form. HPRA [24] overcomes the k -uniform limitation by determining new hyperedge cardinality from the hyperedge degree distribution, then uses the Node-Hyperedge Attachment Score (NHAS) to sequentially select nodes until the hyperedge is complete.

Despite the existence of these methods, they primarily focus on higher-order relations in homogeneous networks, with limited expressive power that cannot be applied to HINs. Furthermore, most hyperedge prediction methods only determine whether a hyperedge will form among a group of nodes, without revealing the internal relations. In contrast, our approach seeks to uncover these internal relations, making our objective distinct from those methods. Therefore, our paper primarily focuses on predicting higher-order relations on HINs, which is still in the preliminary exploration stage.

2.2. Heterogeneous Network Representation Learning

Heterogeneous Network Representation Learning (HNRL) maps diverse nodes into a continuous low-dimensional vector space, where similar nodes are closer while dissimilar nodes are farther apart. It is an effective approach for network analysis, enabling relation prediction based on learned embeddings. The quality of learned embeddings determines the effectiveness of relation prediction. Currently, HNRL can generally be divided into two categories [26]: proximity preservation [27–32] and message-passing methods [7, 33–37]. Proximity preservation methods learn network structure by capturing direct or indirect proximity between nodes. For example, metapath2vec [27]

captures contextual structure through random walks, while NSHE [28] preserves pairwise proximity and network schema structure.

Message-passing methods are the mainstream approach in recent times, learning to generate embeddings for target nodes by aggregating information from their neighbors. They can be roughly categorized into metapath-based methods and relation-based methods [38]. Metapath-based methods mainly preserve heterogeneity by constructing metapath-based neighbors. For example, MAGNN [33] utilizes intra-path aggregation and inter-path aggregation to generate node embeddings. HMSG [36] captures structural, semantic, and attribute information from metapath-based homogeneous and heterogeneous neighbors. Relation-based methods do not require manually setting meta-paths, and directly aggregate the features of one-hop neighbors to learn node representations. For instance, HetSANN [35] employs a type-aware attention layer and embeds nodes by different types of neighbor nodes and associated edges. PHGNN [34] hierarchically aggregates multi-type information using two mechanisms: content-level learning and type-level learning. However, these HNRL methods primarily focus on pairwise relation proximity, often yielding suboptimal results for higher-order relations involving a group of nodes, which will be proved in our experiments.

2.3. Comparison

Compared to previous research on heterogeneous networks, the innovation of this paper lies in:

- (1) **Target Objects:** On one hand, existing embedding-based methods rely on metapaths or motifs as prior knowledge to guide node representation learning, where metapaths or motifs serve as a guiding framework to enhance the representation in downstream tasks, achieving good performance in node-level and graph-level tasks. In contrast, our method directly investigates higher-order relations, expanding the research focus from individual nodes to multiple nodes. On the other hand, although hyperedges represent a form of higher-order relations, they do not uncover the intricate patterns of interactions within a specific set of nodes. Hyperedge-based methods generally focus more on the collective interactions at the event level, neglecting the detailed patterns of interactions among individual nodes within those events, so they lack flexibility in predicting the relation among any given set of nodes. We focus on more specific local structures, whereas hyperedges are unable to distinguish detailed internal structures and have a broader predictive scope.
- (2) **Tasks:** Traditional methods are typically designed for predicting edges between pairs of nodes. In contrast, our approach extends the traditional link prediction task to simultaneously predict relationships among a group of nodes, even including arbitrary heterogeneous type relations.
- (3) **Flexibility:** Regardless of whether its meta-paths or meta-graphs, their relational patterns are predefined or constrained, making them unable to adaptively learn and predict arbitrary heterogeneous relationships. In contrast, our method includes any type of relationship within higher-order relationships, without such limitations.

In summary, compared to previous methods, our model expands from low-order to higher-order, from homogeneous to heterogeneous, and from specific types of relationships to arbitrary heterogeneous types. This broadens the coverage significantly and enhances the network's learning capability.

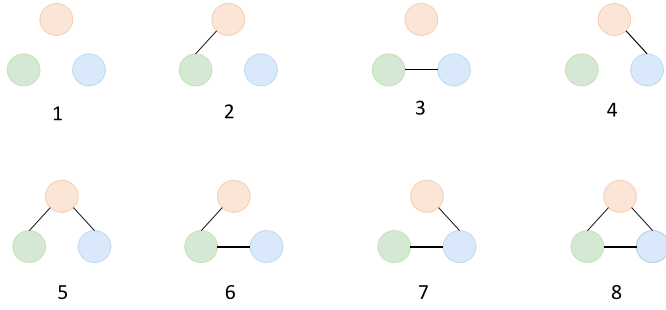


Fig. 2. All the 3-node higher-order relation patterns of the example HIN in Fig. 1 and different colors represent different types of nodes.

3. Preliminaries

In this section, we give formal definitions of some important concepts related to this paper.

Definition 3.1 (Heterogeneous Information Network). A Heterogeneous Information Network (HIN) is represented as $G = (\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R})$ where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of nodes and edges. The node type mapping function satisfies $\phi : \mathcal{V} \rightarrow \mathcal{A}$ and the edge type mapping function satisfies $\psi : \mathcal{E} \rightarrow \mathcal{R}$, and $|\mathcal{A}| + |\mathcal{R}| > 2$. Besides, $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times F}$ is the node feature matrix, and $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ is the adjacency matrix, where F is the dimension of features.

Definition 3.2 (Heterogeneous Higher-Order Relation). A higher-order relation of the HIN G is defined as $c = (\mathcal{V}_c, \mathcal{E}_c, \mathcal{A}_c, \mathcal{R}_c)$, where $\mathcal{V}_c \in \mathcal{V}$ and $|\mathcal{V}_c| > 2$. It satisfies the injective map $\gamma : \mathcal{V}_c \rightarrow \mathcal{V}$, and $(i, j) \in \mathcal{E}_c$ iff $(\gamma(i), \gamma(j)) \in \mathcal{E}$. $\mathcal{A}_c$ and $\mathcal{R}_c$ are the type sets of $\mathcal{V}_c$ and $\mathcal{E}_c$.

Definition 3.3 (Heterogeneous Higher-Order Relation Pattern). Each higher-order relation c corresponds to a pattern, which is represented as $P_c = (\mathcal{A}_c, \mathcal{R}_c)$. As shown in Fig. 2, we take the 3-node heterogeneous higher-order relation as an example, where different colors represent different types of nodes. The 3-node pattern set $\mathcal{P}^3$ contains eight distinct patterns formed by different combinations of edges. For each 3-node heterogeneous higher-order relation c , $P_c \in \mathcal{P}^3$.

For ease of description and comprehension, and without loss of generality, this paper focuses on 3-node heterogeneous higher-order relations where all node types are different. It is worth noting that our approach can be easily extended to more intricate higher-order relations.

Definition 3.4 (Higher-order relation prediction task). Given the observed HIN G (the network may be incomplete) and its higher-order relation set $\mathcal{C}$, our objective is to learn a function f , for each $c \in \mathcal{C}$, making $f(c) = P_c$.

4. Method

In this section, we describe a novel model, ANSE-HGN, designed for predicting higher-order relations in HINs. Fig. 3 illustrates the entire framework of the ANSE-HGN model. In 4.1, we introduce the two primary encoders used in this paper. Section 4.2 details the decoder employed for higher-order relationship prediction. In 4.3, we describe the loss functions that our method utilizes. Finally, in Section 4.4, we provide an analysis of the model's complexity.

4.1. Encoder

In this part, we provide a detailed overview of the two primary encoders used in this paper: the relation-based heterogeneous graph

neural network, which encodes the fundamental structure of the network, and the structure feature encoder, which is designed to accurately capture higher-order proximities. These two encoders enhance the ability to capture complex higher-order relations in heterogeneous networks. The details are introduced in the following subsections.

4.1.1. Heterogeneous graph neural network

We first employ a HGNN to learn the fundamental structure and attributes of the network. For metapath-based HGNNs, it is common practice to utilize meta-paths to decompose the HIN into multiple homogeneous networks, which are then encoded using multiple GNNs. However, defining meta-paths typically requires prior knowledge, and the network decomposition results in each subgraph retaining only one type of node. To avoid these issues, we propose leveraging the relation-based HGNN directly for learning on HINs. In the k th layer, for node i with type t , its neighbors $\mathcal{G}_i^r$ that are based on relation $r \in \mathcal{R}$ is represented as follows:

$$\mathbf{e}_r^k(i) = \text{ReLU}(\mathbf{W}_r^k \cdot \text{AGG}(\mathbf{h}_j^{k-1} | j \in \mathcal{G}_i^r)), \quad (1)$$

where $\mathbf{W}_r^k$ is the weight matrix for relation r in the k th layer, $\mathbf{h}_j^{k-1}$ represents the node representation of neighbor j based on relation r ($\mathbf{h}_j^0$ is the initial representation $\mathbf{x}_j$) in the $(k-1)$ -th layer and $\text{AGG}(\cdot)$ denotes the aggregation function, which can have multiple choices, such as sum aggregation, pooling methods, mean aggregation, and others. Here we use the mean aggregation function to replace it, which is not only insensitive to the number of neighbor nodes but also highly efficient.

In HINs, each node has various types of neighbors based on different relations. To capture all neighbor information, we need to integrate them of different types. We employ the attention mechanism to learn weights for different relations, thereby obtaining better representations. For node i with type t , the attention coefficients s_r^k for relation r is calculated as follows:

$$s_r^k = \frac{1}{|\mathcal{V}^t|} \sum_{i \in \mathcal{V}^t} \tanh((\mathbf{q}_t^k)^T (\mathbf{W}_t^k \cdot \mathbf{e}_r^k(i) + \mathbf{b}_t^k)), \quad (2)$$

Where $\mathbf{W}_t^k$ is weight matrix of type t in the k th layer, $\mathcal{V}^t$ denotes nodes with type t , $\mathbf{b}_t^k$ is bias vector, and $\mathbf{q}_t^k$ is attention vector. Then we normalize the attention coefficients for all relations $\mathcal{R}_t$ associated with node type t :

$$\alpha_r^k = \frac{\exp(s_r^k)}{\sum_{r \in \mathcal{R}_t} \exp(s_r^k)}. \quad (3)$$

After that, we aggregate relation-based neighbors to obtain the neighborhood representation of node i in the k th layer:

$$\mathbf{h}_i^k = \sum_{r \in \mathcal{R}_t} \alpha_r^k \cdot \mathbf{e}_r^k(i). \quad (4)$$

Finally, we merge the learned neighborhood representation of node i after K layers with its original attributes $\mathbf{x}_i$ ($\mathbf{h}_i^0$) to obtain the final representation:

$$\mathbf{h}_i = \mathbf{W}_t \cdot (\mathbf{h}_i^K \parallel \mathbf{x}_i), \quad (5)$$

where $\mathbf{W}_t$ is the parameter matrix for node type t and $(\cdot \parallel \cdot)$ denotes the concatenation of two vectors. Therefore, the embeddings $\mathbf{H} \in \mathbb{R}^{|\mathcal{V}| \times F}$ of all nodes serve as the base embeddings of the network structure and attributes.

4.1.2. Structure feature encoder

Due to the issues demonstrated by HGNNs in Fig. 1(b), to effectively capture the proximity in heterogeneous higher-order relations, we introduce the structure feature encoder. It mainly comprises the relation-based structure encoder, neighbor-based structure encoder, and similarity computation. Firstly, considering the heterogeneity of the network, we utilize independent encoders for different types of edges in HINs. For edge (i, j) , its structure feature value d_{ij} is:

$$d_{ij} = f_r(a_{ij}), \quad (6)$$

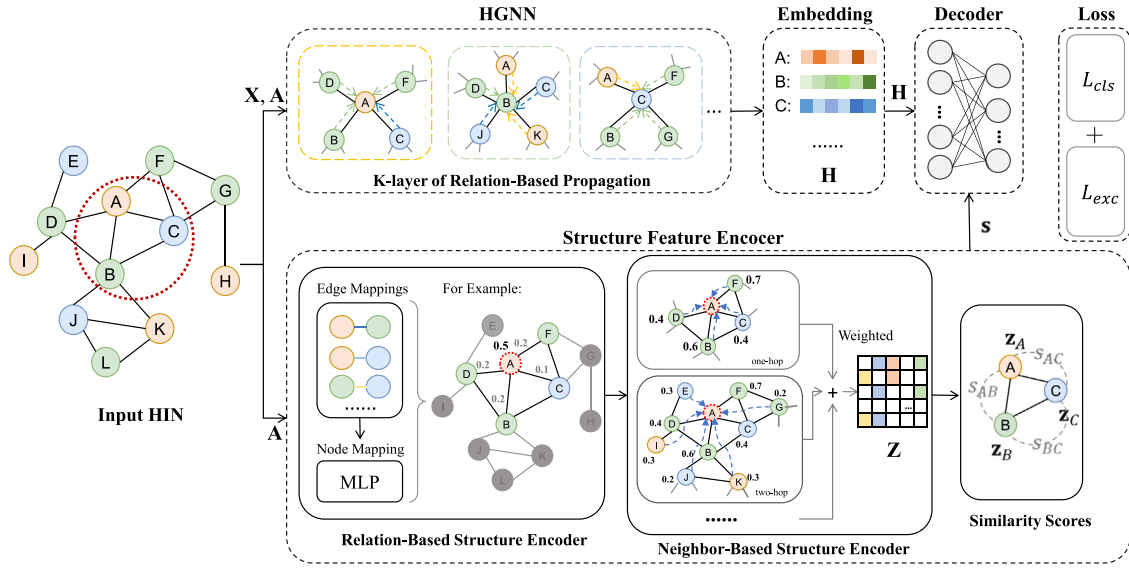


Fig. 3. The overall architecture of ANSE-HGN.

where a_{ij} is the entry of edge (i, j) in the adjacency matrix A . $f_r(\cdot)$ is the relation mapping function for edge type r and we instantiate it using a Multi-Layer Perceptron (MLP).

Afterward, we obtain node structure feature values by aggregating their associated edge features. For node i ,

$$u_i = g(\text{AGG}(d_{ij}) | j \in G_i), \quad (7)$$

where u_i is the structure feature value of node i . G_i is the one-hop neighbors of node i . $\text{AGG}(\cdot)$ is an aggregation function, and here we use a summation function to replace it. $g(\cdot)$ is the node mapping function and we instantiate it using a MLP.

The neighborhood structure plays a crucial role in determining whether relations form. If there is an overlap in the neighborhoods of nodes, the overlapping parts often serve as factors leading to relation formation, which also represents the proximity between the nodes. Consequently, the feature vector similarity between nodes should be higher. After obtaining the structure feature values of all nodes, we generate the neighborhood structure feature vector for each node to characterize its neighborhood. To preserve information and ensure distinctiveness, we vectorize each node's structure feature value with the dimensionality equal to the number of nodes $|V|$. This vectorization preserves the feature values along the dimension corresponding to the node while setting all other dimensions to zero, thus resulting in the structure feature matrix $U \in \mathbb{R}^{|V| \times |V|}$, where each row represents the structure feature vector of the corresponding node.

Therefore, the neighborhood feature matrix of the m -hop neighborhood would be:

$$Z = q\left(\sum_{m=1}^M A^m U\right), \quad (8)$$

where $q(\cdot)$ is a scale controller. Specifically, when $M=1$, Z represents only the one-hop neighborhood, where each row of Z contains all the one-hop neighbor feature values corresponding to the respective node. If we only consider the overlap of the one-hop neighborhood, then we focus solely on whether there are common direct neighbors. However, if we consider the overlap of multi-hop neighborhoods, we assess a broader range of overlapping areas, determining whether there are common indirect neighbors. This broader assessment can reveal important factors that may lead to relation formation. Additionally, weights can be assigned to each hop neighborhood to retain more useful information, as neighbors farther away from the source node have less influence.

Finally, we jointly calculate the similarity for all relations in higher-order relations to determine their neighborhood overlap. For relation (i, j) , the similarity score is computed as follows:

$$s_{ij} = \text{sim}(z_i, z_j), \quad (9)$$

where $\text{sim}(\cdot, \cdot)$ is the function with many choices, we use the inner product. Therefore, for higher-order relation $c = \{i, j, o\}$, its structural similarity can be represented using concatenation as s_c :

$$s_c = [s_{ij} \| s_{io} \| s_{jo}]. \quad (10)$$

4.2. Decoder

The objective of the decoder is to predict higher-order relation patterns. We use both base embeddings and structural similarity as inputs and then predict patterns with a MLP to construct a multi-class decoder. For the higher-order relation $c = \{i, j, o\}$, its predicted pattern is represented as:

$$\hat{y} = \text{MLP}((h_i \| h_j \| h_o) \| s_c). \quad (11)$$

4.3. Loss

Since our task predicts real higher-order relation patterns, which belong to a multi-classification task, we utilize cross-entropy loss for optimization:

$$L_{cls} = - \sum_{c \in C} \sum_{w=1}^W y_c^w \log(\hat{y}_c^w), \quad (12)$$

where W is the number of classes.

Additionally, we consider that leveraging known higher-order relation information (self-information) can help eliminate some impossible real patterns. For example, if a higher-order relation is observed in the network as pattern 2 in Fig. 2, then the actual true pattern cannot be 1, 3, 4, and 7, because these patterns do not contain the edges already in pattern 2. Therefore, we can exclude these patterns. Furthermore, among these patterns that need to be excluded, the less connected patterns should have higher weights, as weaker connectivity implies specific structural deficiencies or functional disruptions, making them more likely to be excluded. In addition to connectivity, if the central node has fewer edges, the weight of the pattern should be higher, as central nodes are expected to have important connections and interactions with other nodes. In such cases, it implies that the pattern is

Algorithm 1 The learning procedure of ANSE-HGN

Require: The HIN G , the initial embeddings $\mathbf{X}$, the higher-order relations set C , the label Y of higher-order relations C , the epoch number E and the neighborhood hops parameter M .

Ensure: Model parameters Θ .

- 1: Initialize parameters Θ ;
- 2: Preprocess to obtain the excluded patterns B of each higher-order relation c .
- 3: **for** $epoch = 1$ to E **do**
- 4: Encode the HIN G with a relation-based HGNN as embeddings $\mathbf{H}$ via Eq. (1)-(5);
- 5: Generate structure feature value $\{u_i\}_{i \in \mathcal{V}}$ via Eq. (6)-(7);
- 6: Characterize M -hop neighborhood as $\mathbf{Z}$ for all nodes via Eq. (8);
- 7: Calculate the similarity of the neighborhood via Eq. (9);
- 8: Decode patterns of higher-order relations C via Eq. (11);
- 9: Update Θ via Eq. (12)-(16);
- 10: **end for**
- 11: **return** Model parameters Θ ;

more dispersed and less likely. Therefore, we design a weight function for each impossible pattern b as follows:

$$\omega(b) = e^{-(n(b)+p(b))}, \quad (13)$$

where $n(b)$ is the connectivity of b , we use the number of edges in the pattern b as the value for $n(b)$. $p(b)$ is the centrality of b , we set its value equal to the number of edges connected to the central node.

Based on the above idea, we design an auxiliary loss. Assuming B denotes all impossible patterns for the higher-order relation c , we use binary classification loss to calculate the loss for all impossible patterns:

$$l_{exc}(c) = - \sum_{b \in B} \omega(b) \cdot (\tilde{y} \cdot \log(\tilde{y}_c^b) + (1 - \tilde{y}) \cdot \log(1 - \tilde{y}_c^b)), \quad (14)$$

where $\tilde{y}$ is the label of the excluded patterns, we set it to zero. Therefore, for all higher-order relations C , the loss is:

$$L_{exc} = \sum_{c \in C} l_{exc}(c). \quad (15)$$

Overall, the total training loss of our model is:

$$L = L_{cls} + \beta L_{exc}, \quad (16)$$

where β is the hyperparameter used to balance the two parts of the loss. The learning procedure of the model is described in Algorithm 1.

4.4. Complexity analysis

In a relation-based HGNN, the time complexity of each layer typically depends on the aggregation of node features and the computation of weights using the attention mechanism. The total complexity is $O(|\mathcal{V}|F' + |\mathcal{E}|F')$, where F is the input dimension and F' is the output dimension. In the structure feature encoder, the multiplication of the adjacency matrix can be precomputed without affecting the training time. Additionally, we use a structure feature matrix $\mathbf{U}$ to represent the structural features of each node. The complexity of computing the neighborhood feature representation $\mathbf{Z}$ is $O(|\mathcal{V}| \|\mathcal{E}\|_M)$, where $|\mathcal{E}|_M$ denotes the number of edges connecting the M -hop neighborhood. To improve computational efficiency, we use sparse matrices to represent them in practice, thus the time complexity becomes $O(|\mathcal{E}|_M)$. For the decoder, the time complexity is $O(|C|F'H)$, where $|C|$ is the number of higher-order relations, H is the number of hidden layer units.

5. Experiments

In this section, we first describe the experimental setup, including the datasets, baseline methods, parameter settings, and evaluation metrics. We then present the comparative experimental results on the higher-order relation prediction task, benchmarking our method against baseline methods.

5.1. Datasets

We evaluate our model on four real heterogeneous datasets, which are described in detail in Table 1.

- (1) **DBLP**¹: This is a bibliographic website for computer science. We construct a HIN involving authors (A), papers (P), conferences (C), and terms (T). The target node types used to construct 3-node higher-order relations include authors, papers, and conferences.
- (2) **DouBan**²: The DouBan website contains information about books. We use books (B), authors (A), and publishers (P) to construct a HIN and extract 3-node higher-order relations for prediction.
- (3) **Aminer**³: This is a website that provides a plethora of academic information. We use papers (P), authors (A), and conferences (C) to construct a HIN and extract 3-node higher-order relations for prediction.
- (4) **IMDB**⁴: This is an online database and website that contains information about movies. We use users (U), movies (M), ratings (R), and genres (G) to construct a HIN. The target node types used to construct 3-node higher-order relations include users, movies, and genres.

5.2. Baselines

Given the inadequacy of current methods in addressing the prediction of higher-order relations in HINs, we broaden our analysis by extending two categories of baseline methods for comparative evaluation. The first category includes embedding-based models such as DeepWalk [39], Metapath2vec [27], NSHE [28], RHINE [40], HIN2vec [41], GATNE [42], and HeGAN [43]. For these methods, we concatenate the embeddings of multiple nodes within higher-order relations to depict the relations. Subsequently, we employ a MLP classifier for prediction. The second category includes heterogeneous link prediction models such as HGT [7], SimpleHGN [8], SLICE [44], NOH [45] and HeteHGVAE [46]. We predict links separately for each pair of nodes within the higher-order relations and then combine all links together as the final prediction pattern for higher-order relations.

- **DeepWalk** [39] is a classic method for homogeneous network embedding. It leverages random walk sequences, followed by the application of a neural language model to derive network embeddings.
- **Metapath2vec** [27] extends DeepWalk method and is specifically tailored for HINs. It employs metapath-based walks to effectively capture relations between various types of nodes and introduces heterogeneous Skip-gram to account for node types within the network structure.
- **RHINE** [40] divides the relations in HINs into one-centered-by-another relations and peer-to-peer relations, then establishes different models to perceive the relation structure and learn node representations.
- **HIN2vec** [41] employs random walks and negative sampling to generate training data, predicting whether two nodes possess a particular relation, thus effectively learning embeddings of nodes and meta-paths.
- **GATNE** [42] prioritizes distinguishing between various relations among node pairs, and supports both transductive and inductive learning for embedding each node across different types of edges.

¹ <https://github.com/cynricfu/MAGNN>

² <https://github.com/librahu/HIN-Datasets-for-Recommendation-and-Network-Embedding>

³ <https://www.aminer.cn/citation>

⁴ <https://grouplens.org/datasets/movielens/100k/>

Table 1
Summary of datasets.

Dataset	#Nodes	#Edges	#Higher-order relations
DBLP	A: 14,475; P: 14,376; C: 20; T: 8,920	P-A: 41,794; P-C: 14,376; P-T: 114,624; A-C: 24,495	8,750
DouBan	A: 2,141; B: 3,000; P: 615	A-B: 3,000; B-P: 3,000; A-P: 2,531	8,590
Aminer	A: 13,143; P: 3,000; C: 280	P-A: 14,331; P-C: 5,000; A-A: 19,165; A-C: 13,663	8,529
IMDB	U: 943; M: 1,682; R: 5; G: 19	U-M: 100,000; U-R: 4,423; M-G: 2,893; M-R: 6,714; U-G: 14,236	14,000

Table 2

Experiment results on the DouBan and DBLP datasets measured by Macro-F1, Micro-F1, NMI for the higher-order relation prediction task. The best and the suboptimal performances are marked in bold and underlined respectively.

Method	DouBan			DBLP		
	Macro-F1	Micro-F1	NMI	Macro-F1	Micro-F1	NMI
Deepwalk	0.424 $\pm$ 0.053	0.462 $\pm$ 0.061	0.561 $\pm$ 0.070	0.380 $\pm$ 0.026	0.443 $\pm$ 0.011	0.447 $\pm$ 0.050
Metapath2vec	0.484 $\pm$ 0.053	0.482 $\pm$ 0.034	0.536 $\pm$ 0.068	0.343 $\pm$ 0.136	0.403 $\pm$ 0.110	0.352 $\pm$ 0.105
RHINE	0.232 $\pm$ 0.161	0.317 $\pm$ 0.150	0.294 $\pm$ 0.255	0.155 $\pm$ 0.084	0.248 $\pm$ 0.087	0.228 $\pm$ 0.142
HeGAN	0.510 $\pm$ 0.072	0.502 $\pm$ 0.070	0.424 $\pm$ 0.027	0.492 $\pm$ 0.014	0.502 $\pm$ 0.010	0.457 $\pm$ 0.007
HIN2vec	0.144 $\pm$ 0.057	0.239 $\pm$ 0.058	0.259 $\pm$ 0.076	0.151 $\pm$ 0.046	0.246 $\pm$ 0.035	0.199 $\pm$ 0.054
GATNE	0.411 $\pm$ 0.165	0.468 $\pm$ 0.149	0.523 $\pm$ 0.214	0.271 $\pm$ 0.098	0.363 $\pm$ 0.085	0.402 $\pm$ 0.126
NSHE	0.331 $\pm$ 0.110	0.406 $\pm$ 0.115	0.463 $\pm$ 0.134	0.286 $\pm$ 0.089	0.383 $\pm$ 0.074	0.351 $\pm$ 0.092
HGT	0.594 $\pm$ 0.012	0.622 $\pm$ 0.007	0.652 $\pm$ 0.012	0.519 $\pm$ 0.030	0.548 $\pm$ 0.023	0.511 $\pm$ 0.021
SLiCE	0.111 $\pm$ 0.012	0.130 $\pm$ 0.012	0.039 $\pm$ 0.011	0.110 $\pm$ 0.007	0.126 $\pm$ 0.005	0.149 $\pm$ 0.011
SimpleHGN	0.369 $\pm$ 0.051	0.459 $\pm$ 0.046	0.322 $\pm$ 0.041	0.071 $\pm$ 0.006	0.124 $\pm$ 0.022	0.324 $\pm$ 0.083
NOH	0.492 $\pm$ 0.036	0.482 $\pm$ 0.035	0.409 $\pm$ 0.017	0.374 $\pm$ 0.028	0.436 $\pm$ 0.014	0.460 $\pm$ 0.065
HeteHG-VAE	0.477 $\pm$ 0.086	0.490 $\pm$ 0.062	0.444 $\pm$ 0.039	0.462 $\pm$ 0.011	0.515 $\pm$ 0.014	0.456 $\pm$ 0.007
ANSE-HGN	0.687 $\pm$ 0.041	0.723 $\pm$ 0.044	0.729 $\pm$ 0.048	0.593 $\pm$ 0.056	0.600 $\pm$ 0.057	<u>0.505 $\pm$ 0.042</u>

- **HeGAN** [43] is the first to utilize GAN in HNRL. By incorporating multiple relations into both the generator and discriminator that are trained through a minimax game, it can account for the heterogeneity.
- **NSHE** [28] proposes a network schema sampling method to address the bias among various node types and utilizes multi-task learning to preserve both pairwise structure and network schema structure.
- **HGT** [7] utilizes meta relation parameterized weight matrices, allowing different types of nodes and edges to maintain their specific representation spaces, thereby learning the heterogeneity of the network.
- **SimpleHGN** [8] incorporates GAT as the backbone with improvements stemming from the refinement of three established methodologies: edge-type embedding, residual connections, and L2 normalization.
- **SLiCE** [44] is a novel self-supervised learning approach that grasps higher-order semantic relationships among nodes by concurrently capturing global and local aspects of context subgraphs.
- **NOH** [45] captures higher-order relationships between nodes by constructing a heterogeneous hypergraph. It utilizes a variational autoencoder for information propagation and aggregation and employs a neighborhood overlap-aware graph neural network to generate node representations enriched with structural information.
- **HeteHG-VAE** [46] maps a heterogeneous information network to a heterogeneous hypergraph, using a Bayesian deep generative framework and hyperedge attention to capture higher-order relationships and learn latent representations of nodes and hyperedges.

5.3. Setup

We construct the heterogeneous network by randomly removing 20% of the edges from the original HIN. Then we extract heterogeneous

higher-order relations for prediction, of which the number is given in Table 1. It is worth noting that, due to the nature of the data, the extracted higher-order relations do not include pattern 5 shown in Fig. 2. For instance, in academic networks, if an author has published a paper at a conference, there will be a connection between the author and the conference, thus resulting in the absence of pattern 5. To ensure data balance, we extract an equal number of higher-order relations for other patterns. We divide the dataset into training, validation, and testing sets in a ratio of 3:1:1. The experiments are conducted in an Ubuntu 18.04 environment, utilizing an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. The implementation was carried out using Python 3.6, with the deep learning models developed and trained using the PyTorch framework. More details of ANSE-HGN are available at <https://anonymous.4open.science/r/ANSE-HGN-3378/>.

For our model, the training epoch number is 500, with the embedding dimension set to 64, utilizing the Adam optimizer [47] with a learning rate of $3e-4$. The parameters for M hops are set to 2 in DouBan and AMiner, and 1 in IMDB and DBLP, β is set to 1, and the number of HGNN layers K is set to 2. We initialize node features using the DeepWalk [39] method.

For baseline that use meta-paths in modeling, we predefined meta-paths on the four datasets as APCPA, ABPBA, APCPA, and UMGMU, respectively. For models that use random walk, the number of walks is 200, the walk length is 50, and the window size is 5. We attempt to follow the settings described in the original papers regarding other parameters.

In the experiments on higher-order relation prediction, we include two tasks: classification and clustering. For classification, we evaluate using Macro-F1 and Micro-F1 scores, while for clustering, we use Normalized Mutual Information (NMI) to assess the similarity between the clustering results and the ground truth labels. We report the mean $\pm$ standard deviation across ten seeds for all methods.

5.4. Higher-order relation prediction results

We conduct higher-order relation prediction tasks on four datasets. Tables 2 and 3 provides a comparison of our model with baselines.

Table 3

Experiment results on the Aminer and IMDB datasets measured by Macro-F1, Micro-F1, NMI for the higher-order relation prediction task. The best and the suboptimal performances are marked in bold and underlined respectively.

Method	Aminer			IMDB		
	Macro-F1	Micro-F1	NMI	Macro-F1	Micro-F1	NMI
Deepwalk	<u>0.640 ± 0.017</u>	<u>0.689 ± 0.016</u>	<u>0.765 ± 0.015</u>	0.417 ± 0.028	0.476 ± 0.017	0.703 ± 0.036
Metapath2vec	0.438 ± 0.050	0.507 ± 0.031	0.614 ± 0.023	0.412 ± 0.016	0.447 ± 0.015	0.685 ± 0.032
RHINE	0.060 ± 0.032	0.160 ± 0.026	0.048 ± 0.070	0.228 ± 0.080	0.289 ± 0.047	0.505 ± 0.155
HeGAN	0.599 ± 0.017	0.644 ± 0.018	0.694 ± 0.018	0.417 ± 0.026	0.453 ± 0.022	<u>0.703 ± 0.025</u>
HIN2vec	0.347 ± 0.075	0.419 ± 0.057	0.510 ± 0.072	0.268 ± 0.088	0.433 ± 0.091	0.392 ± 0.133
GATNE	0.635 ± 0.014	0.689 ± 0.007	0.753 ± 0.006	<u>0.474 ± 0.118</u>	0.536 ± 0.066	0.595 ± 0.067
NSHE	0.566 ± 0.134	0.623 ± 0.122	0.686 ± 0.124	0.472 ± 0.015	<u>0.538 ± 0.011</u>	0.556 ± 0.010
HGT	0.346 ± 0.070	0.424 ± 0.081	0.400 ± 0.070	0.426 ± 0.064	0.507 ± 0.039	0.654 ± 0.027
SLiCE	0.106 ± 0.010	0.122 ± 0.009	0.054 ± 0.024	0.105 ± 0.011	0.119 ± 0.013	0.049 ± 0.008
SimpleHGN	0.346 ± 0.070	0.424 ± 0.081	0.400 ± 0.070	0.071 ± 0.006	0.124 ± 0.022	0.324 ± 0.083
NOH	0.319 ± 0.023	0.360 ± 0.031	0.454 ± 0.053	0.356 ± 0.175	0.415 ± 0.133	0.327 ± 0.111
HeteHGN-VAE	0.284 ± 0.097	0.348 ± 0.080	0.346 ± 0.137	0.510 ± 0.050	0.516 ± 0.041	0.513 ± 0.058
ANSE-HGN	0.719 ± 0.070	0.741 ± 0.060	0.765 ± 0.025	0.608 ± 0.024	0.653 ± 0.018	0.679 ± 0.037

Table 4

Ablation results on the DouBan dataset measured by Macro-F1, Micro-F1, NMI for the higher-order relation prediction task. The best performances are marked in bold.

DouBan	Macro_F1	Micro_F1	NMI
ANSE-HGN	0.687 ± 0.041	0.723 ± 0.044	0.729 ± 0.048
ANSE-HGN w/o Struct	0.658 ± 0.005	0.689 ± 0.006	0.682 ± 0.006
ANSE-HGN w/o L_{exc}	0.663 ± 0.095	0.703 ± 0.088	0.709 ± 0.091
ANSE-HGN w/o (Struct+ L_{exc})	0.650 ± 0.009	0.682 ± 0.009	0.668 ± 0.012
ANSE-HGN w/o (HGNN+ L_{exc})	0.653 ± 0.189	0.670 ± 0.168	0.600 ± 0.162

The best and the suboptimal performances are marked in bold and underlined respectively.

The results indicate that ANSE-HGN is significantly superior to all baselines. In the DouBan and DBLP datasets, HGT achieved suboptimal results, possibly because it does not parameterize each type of edge but instead decomposes each edge based on detailed meta-relation triplets. However, since it only focuses on pairwise nodes and cannot capture the proximity in higher-order relations, its performance is lower than our model. The same is true for SLiCE and SimpleHGN. On the Aminer and IMDB datasets, random walk based methods like DeepWalk have shown promising results, this demonstrates that random walk is an effective approach for capturing higher-order proximity. Metapath2vec and HIN2vec have shown mediocre results, possibly due to their reliance on meta-paths, limiting their ability to capture direct neighbor proximity. HeGAN, leveraging adversarial learning, also achieved favorable results. In contrast, RHINE's simplistic models without deep neural networks lead to relatively poor performance, making it challenging to capture higher-order proximity in complex networks. However, these methods only retain lower-order proximity and achieve suboptimal performance in predicting higher-order relations compared to our model. Although HGT achieved good results in link prediction, the formation of the network does not occur pairwise. Predicting relations one by one and then combining them may fail to capture the essence of higher-order relations. A single prediction error could lead to the collapse of the entire higher-order relation prediction. This highlights the superior capability and effectiveness of ANSE-HGN in handling higher-order relations. The NOH and HeteHG-VAE methods, which utilize hyperedges, significantly underperform compared to our model. This is because they focus only on the relationship with the central node when considering hypergraphs, and do not take into account the relationships between non-central nodes. In contrast, our model for higher-order relation prediction more comprehensively covers all heterogeneous relationships (whether central or non-central nodes). Hence they fail to capture the proximity of non-central node connections, which limits their effectiveness in higher-order relation prediction. However, Our encoder is not restricted in this manner.

5.5. Ablation study

To analyze the importance of each component in ANSE-HGN, we conduct an ablation study where we define variants of ANSE-HGN as follows:

- **ANSE-HGN w/o Struct:** ANSE-HGN removes the structure feature encoder, using only the HGNN encoder and the loss L for learning.
- **ANSE-HGN w/o L_{exc} :** ANSE-HGN removes the loss L_{exc} and uses only classification loss L_{cls} for optimization.
- **ANSE-HGN w/o (Struct+ L_{exc}):** ANSE-HGN removes the structure feature encoder and loss L_{exc} , using only the HGNN encoder and the classification loss L_{cls} for optimization.
- **ANSE-HGN w/o (HGNN+ L_{exc}):** ANSE-HGN removes HGNN and loss L_{exc} , only retains the structure feature encoder and loss L_{cls} .

Table 4 displays the results of ablation experiments on the DouBan dataset. We observe a significant decrease in model performance when the structural feature encoder is removed in **ANSE-HGN w/o Struct**, demonstrating its effectiveness in capturing proximity in higher-order relations. Additionally, **ANSE-HGN w/o L_{exc}** proves that our designed self-information loss function provides valuable insights for predictions. It can further boost the model's performance by leveraging the observed edges. The decrease in performance of **ANSE-HGN w/o (Struct+ L_{exc})** similarly indicates that using HGNN alone for higher-order relation prediction is not optimal, thus validating our previous findings regarding the limitations of HGNNs in higher-order relation prediction. Therefore, there is a need to learn more accurate higher-order proximity for higher-order relation modeling. **ANSE-HGN w/o (HGNN+ L_{exc})** shows that omitting HGNN results in a performance decrease, highlighting the necessity of using HGNN as the base embedding. Overall, our model ANSE-HGN achieves optimal performance, significantly contributing to accurately capturing higher-order proximity.

5.6. Visualization

To provide a more intuitive comparison with baseline models, we visualize the embedding results of the higher-order relation prediction

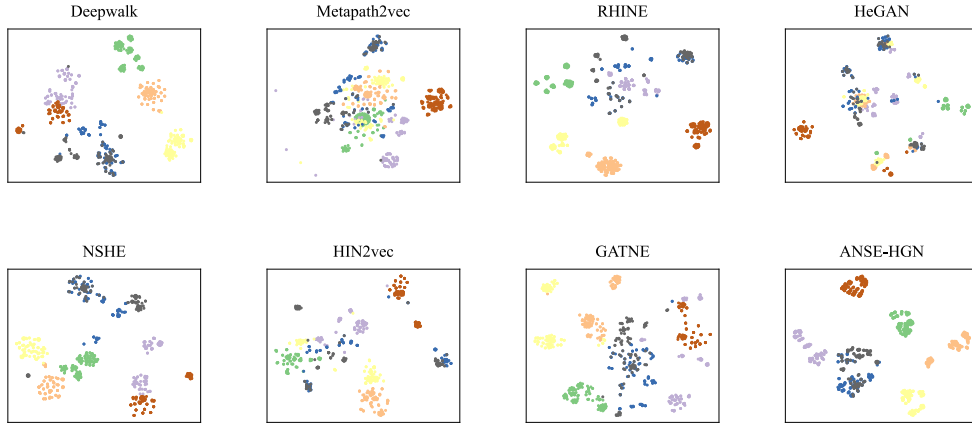


Fig. 4. Embedding visualization of higher-order relation.

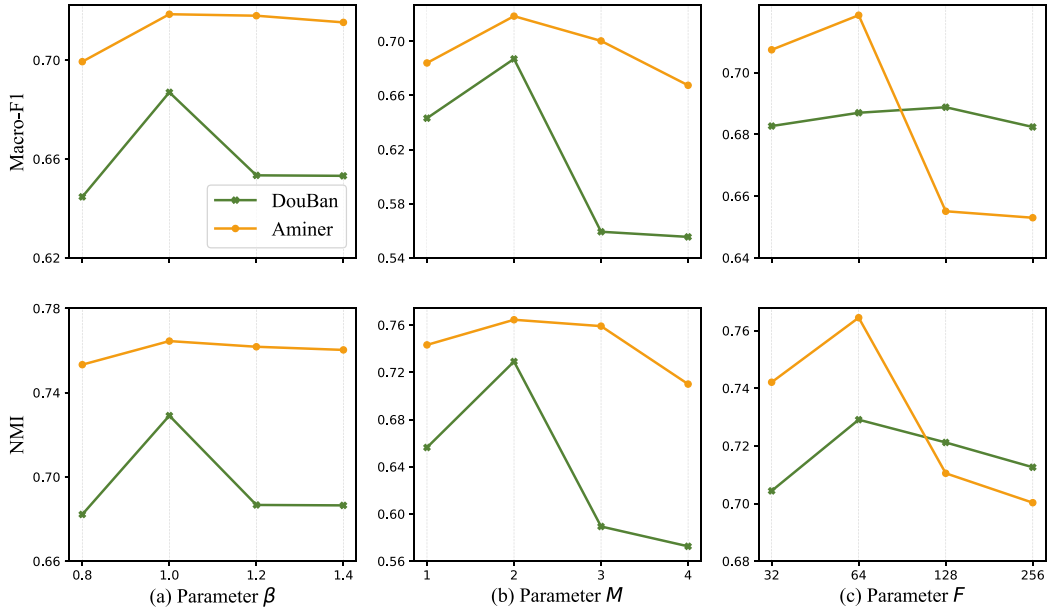


Fig. 5. Parameter analysis of ANSE-HGN on the DouBan and Aminer datasets.

task. Here, we utilize t-SNE to project the learned higher-order relation embeddings (by concatenating the three node base embeddings together) into a two-dimensional space and plot them in Fig. 4. Based on the visualization results, we can observe that in the embedding-based baseline methods like HeGAN, RHINE, and HIN2vec, most categories are mixed with unclear boundaries, potentially leading to inaccuracies in classification tasks. In contrast, ANSE-HGN can separate higher-order relations into different classes with clear boundaries, further demonstrating that our designed modules are more capable of capturing the proximity and distinguishing different higher-order relations.

5.7. Parameter analysis

In this section, we investigate the sensitivity of parameters, including loss weight β , max hop M , and embedding dimension F . Fig. 5 reports the experimental results of ANSE-HGN on the DouBan and Aminer datasets. Fig. 5(a) illustrates the impact of the balance parameter β on different datasets. A larger β indicates a greater influence of self-information on the model. We observe an initial improvement in model performance as balance parameters β increase, peaking at $\beta=1.0$ before declining. Therefore, we set β to 1 in this paper, which represents that self-information loss is as important as classification loss. In Fig. 5(b), the parameter M represents the number of hops

considered in the neighborhood. We observe that initially, as M increases, the performance also improves. This is because the model aggregates more neighborhood information, leading to better performance. However, when M exceeds a certain threshold, the additional distant neighborhood information may provide fewer valuable insights or even introduce noise, resulting in a decrease in the model performance. Therefore, we set M to 2. In Fig. 5(c), different datasets exhibit various optimal results. The embedding dimension F achieves the best performance with a value of 64 on the Aminer dataset, while on the DouBan dataset, dimensions 64 and 128 show promising results. This is because the smaller dimension may not capture enough information about the relations between nodes, while the larger dimension may introduce noise and lead to overfitting. To select a moderate and unified value, we set the dimension uniformly to 64 on all datasets.

We conduct additional experiments on training time for one epoch to compare our method with others. From Fig. 6, we can observe RHINE has the shortest runtime, which can be attributed to its model not utilizing a deep learning framework. This also contributes to its poorer performance. Most models complete an epoch within 100 s, including our model. Simpler models, such as Deepwalk and SimpleHGN, require approximately 10 s or less to train an epoch. HeGAN, SLiCE, and GATNE have runtimes within 300 s. HIN2vec takes the longest due to its code involving multiple loops to generate positive and negative

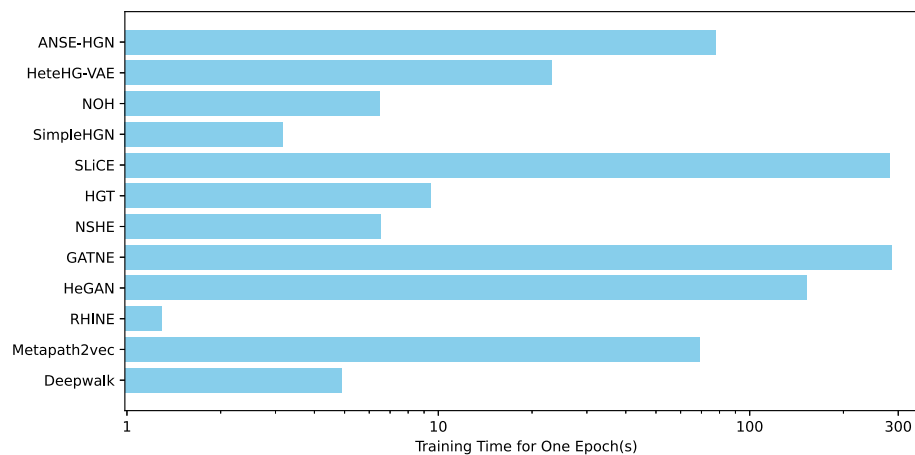


Fig. 6. The comparison of training times per epoch among different models (seconds).

sample pairs. Due to its significantly longer runtime compared to other models, we do not display its results here. The comparison shows that ANSE-HGN performs well in terms of training time. The designed encoder does not add significantly to the time required, making it an effective approach.

5.8. Discussion

This paper aims to predict higher-order relations in heterogeneous networks. Compared to baseline methods, most embedding-based approaches introduce a lot of prior knowledge when modeling networks, such as metapaths and network schemas. However, this can also have negative effects. If the quality of prior knowledge is poor or incomplete, it may lead to overlooking important structures in the network and insufficient learning of non-prior network structures. Our higher-order relations prediction involves relations between any type of nodes, and if the prior knowledge is inadequately defined, it greatly impacts the accuracy of higher-order relation predictions. In contrast, our model does not require any prior knowledge. The structural feature encoder can adaptively learn beneficial structural features, greatly enhancing the model's flexibility in learning network structures. Methods based on link prediction mostly consider the proximity between two nodes, which is difficult to apply in interactions among multiple nodes. Although NOH and HeteHG-VAE attempt to learn higher-order relations through constructing hypergraphs, they focus only on relations with the central node and neglect those among non-central nodes. Therefore, they are unable to capture the proximity of non-central node connections. However, our encoder is not subject to this limitation and can cover all heterogeneous relations, breaking this constraint.

In summary, compared to previous methods, our model expands from low-order to higher-order, from homogeneous to heterogeneous, and from specific types of relationships (connected to the central node) to arbitrary heterogeneous types. This broadens the coverage significantly and enhances the network's learning capability.

6. Conclusion

To address the gap in predicting higher-order relations on HINs, this paper proposes a novel method called ANSE-HGN. ANSE-HGN consists mainly of two encoders: the HGNN encodes the network as the base embedding, and the structure feature encoder captures accurate proximity in higher-order relations using multi-hop neighborhood information. These two kinds of information are then fused for higher-order relation prediction. Additionally, we utilize the self-information of higher-order relationships (connectivity and centrality) to design additional loss functions for optimization. Experimental results on four datasets validate the superiority of our model. In future work, apart

from considering higher-order relations in static networks, we will focus on exploring applications in temporal networks.

This paper marks the initial phase of exploring higher-order relations in heterogeneous networks, with a focus on relations involving three nodes. As the model extends to encompass higher-order relations formed by a larger number of nodes, the number of computations for assessing node similarity increases. Although this feature proves effective, it leads to a dual challenge: not only does the computational burden increase, but the dimensions of the decoder also expand, necessitating the training of the model for higher-order relations of different orders. Therefore, a key area for future improvement we are considering is the possibility of extending the model to any order easily without increasing the computational demands.

CRediT authorship contribution statement

Jie Li: Writing – review & editing, Writing – original draft, Methodology, Data curation. **Xuan Guo:** Investigation, Formal analysis, Conceptualization. **Pengfei Jiao:** Visualization, Validation. **Wenjun Wang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (62472305, 62372146), Key R&D Project of Hainan (ZDYF2024SHFZ051), Zhejiang Provincial Natural Science Foundation of China (LDT23F01015F01).

Data availability

I have shared the link to data in the paper.

References

- [1] W. Zhao, S. Pu, Collaboration prediction in heterogeneous academic network with dynamic structure and topic, *Knowl. Inf. Syst.* 63 (8) (2021) 2053–2074.
- [2] X. Bai, F. Zhang, J. Liu, et al., Scientific Collaboration, Quantifying the impact of and papers via motif-based heterogeneous networks, *J. Informetr.* 17 (2) (2023) 101397.
- [3] Y. Song, X. Yang, C. Xu, Self-supervised calorie-aware heterogeneous graph networks for food recommendation, *ACM Trans. Multimed. Comput. Commun. Appl.* 19 (1s) (2023).

- [4] D. Cai, S. Qian, Q. Fang, et al., User cold-start recommendation via inductive heterogeneous graph neural network, *ACM Trans. Inf. Syst.* 41 (3) (2023) 1–27.
- [5] H. Wang, Z. Cui, R. Liu, et al., A multi-type transferable method for missing link prediction in heterogeneous social networks, *IEEE Trans. Knowl. Data Eng.* 35 (11) (2023) 10981–10991.
- [6] M. Balakrishnan, T.V. G. Network alignment and link prediction using event-based embedding in aligned heterogeneous dynamic social networks, *Appl. Intell.* 53 (20) (2023) 24638–24654.
- [7] Z. Hu, Y. Dong, K. Wang, et al., Heterogeneous graph transformer, in: *Proceedings of the 29th Web Conference*, 2020, pp. 2704–2710.
- [8] Q. Lv, M. Ding, Q. Liu, et al., Are we really making much progress? Revisiting, benchmarking and refining heterogeneous graph neural networks, in: *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2021, pp. 1150–1160.
- [9] T. Nguyen, Z. Liu, Y. Fang, Link prediction on latent heterogeneous graphs, in: *Proceedings of the 32nd Web Conference*, 2023, pp. 263–273.
- [10] Y. Liu, J. Ma, P. Li, Neural predicting higher-order patterns in temporal networks, in: *Proceedings of the 31st Web Conference*, 2022, pp. 1340–1351.
- [11] M. Schaub, A. Benson, P. Horn, et al., Random walks on simplicial complexes and the normalized hodge 1-Laplacian, *SIAM Rev.* 62 (2) (2020) 353–391.
- [12] J. Billings, M. Hu, G. Lerda, et al., Simplex2vec embeddings for community detection in simplicial complexes, 2019, arXiv preprint arXiv:1906.09068.
- [13] U. Alon, Network motifs: Theory and experimental approaches, *Nature Rev. Genet.* 8 (6) (2007) 450–461.
- [14] A. Benson, R. Abebe, M. Schaub, et al., Simplicial closure and higher-order link prediction, in: *Proceedings of the National Academy of Sciences*, vol. 115, (48) 2018, pp. E11221–E11230.
- [15] R. Lambiotte, M. Rosvall, I. Scholtes, From networks to optimal higher-order models of complex systems, *Nat. Phys.* 15 (4) (2019) 313–320.
- [16] A. Benson, D. Gleich, J. Leskovec, Higher-order organization of complex networks, *Science* 353 (6295) (2016) 163–166.
- [17] A. Benson, D. Gleich, D. Higham, Higher-order network analysis takes off, fueled by classical ideas and new data, 2021, arXiv preprint arXiv:2103.05031.
- [18] N. Chavan, K. Potika, Higher-order link prediction using triangle embeddings, in: *The 9th IEEE International Conference on Big Data*, 2020, pp. 4535–4544.
- [19] M. Besta, R. Grob, C. Miglioli, et al., Motif prediction with graph neural networks, in: *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 35–45.
- [20] H. Nassar, A. Benson, D. Gleich, Neighborhood and PageRank methods for pairwise link prediction, *Soc. Netw. Anal. Min.* 10 (1) (2020) 1–13.
- [21] M. Zhang, Z. Cui, S. Jiang, et al., Beyond link prediction: Predicting hyperlinks in adjacency space, in: *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*, 32, (1) 2018, pp. 4430–4437.
- [22] N. Yadati, V. Nitin, M. Nimishakavi, et al., NHP: Neural hypergraph link prediction, in: *Proceedings of the 29th ACM International Conference on Information and Knowledge Management*, 2020, pp. 1705–1714.
- [23] D. Maurya, B. Ravindran, Hyperedge prediction using tensor eigenvalue decomposition, *J. Indian Inst. Sci.* 101 (2021) 443–453.
- [24] T. Kumar, K. Darwin, S. Parthasarathy, et al., HPRA: Hyperedge prediction using resource allocation, in: *Proceedings of the 12th ACM Conference on Web Science*, 2020, pp. 135–143.
- [25] M. Zhang, Z. Cui, M. Neumann, et al., An end-to-end deep learning architecture for graph classification, in: *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*, vol. 32, (1) 2018.
- [26] C. Yang, Y. Xiao, Y. Zhang, et al., Heterogeneous network representation learning: A unified framework with survey and benchmark, *IEEE Trans. Knowl. Data Eng.* 34 (10) (2020) 4854–4873.
- [27] Y. Dong, N. Chawla, A. Swami, Metapath2vec: Scalable representation learning for heterogeneous networks, in: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 133–144.
- [28] J. Zhao, X. Wang, C. Shi, et al., Network schema preserving heterogeneous information network embedding, in: *Proceedings of the 29th International Joint Conference on Artificial Intelligence*, 2020, pp. 1366–1372.
- [29] X. Zhang, Q. Yang, D. Xu, Trans: Transition-based knowledge graph embedding with synthetic relation representation, in: *Findings of the Association for Computational Linguistics*, 2022, pp. 1202–1208.
- [30] W. Zhang, Y. Fang, Z. Liu, et al., Mg2vec: Learning relationship-preserving heterogeneous graph representations via metagraph embedding, *IEEE Trans. Knowl. Data Eng.* 34 (3) (2020) 1317–1329.
- [31] H. Wei, G. Xiong, Q. Wei, et al., Structure-aware attributed heterogeneous network embedding, *Knowl. Inf. Syst.* 65 (4) (2023) 1769–1785.
- [32] S. Forouzandeh, K. Berahm, R. Sheikhpour, et al., A new method for recommendation based on embedding spectral clustering in heterogeneous networks (RESCHet), *Expert Syst. Appl.* 231 (2023) 120699.
- [33] X. Fu, J. Zhang, Z. Meng, et al., MAGNN: Metapath aggregated graph neural network for heterogeneous graph embedding, in: *Proceedings of the 29th Web Conference*, 2020, pp. 2331–2341.
- [34] H. Yang, L. Li, L. Zhang, et al., PHGNN: Position-aware graph neural network for heterogeneous graph embedding, in: *2021 International Joint Conference on Neural Networks*, 2021, pp. 1–8.
- [35] H. Hong, H. Guo, Y. Lin, et al., An attention-based graph neural network for heterogeneous structural learning, in: *Proceedings of the 34th AAAI Conference on Artificial Intelligence*, 34, (04) 2020, pp. 4132–4139.
- [36] M. Guan, X. Cai, J. Shang, et al., HMSG: Heterogeneous graph neural network based on metapath SubGraph learning, *Knowl.-Based Syst.* 279 (2023) 110930.
- [37] S. Forouzandeh, M. Rostami, K. Berahm, et al., Health-aware food recommendation system with dual attention in heterogeneous graphs, *Comput. Biol. Med.* 169 (2024) 107882.
- [38] R. Bing, G. Yuan, M. Zhu, et al., Heterogeneous graph neural networks analysis: A survey of techniques, evaluations and applications, *Artif. Intell. Rev.* 56 (8) (2023) 8003–8042.
- [39] B. Perozzi, R. Al-Rfou, S. Skiena, Deepwalk: Online learning of social representations, in: *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2014, pp. 701–710.
- [40] Y. Lu, C. Shi, L. Hu, et al., Relation structure-aware heterogeneous information network embedding, in: *Proceedings of the 33rd AAAI Conference on Artificial Intelligence*, 33 (01) (2019) 4456–4463.
- [41] T. Fu, W. Lee, Z. Lei, Hin2vec: Explore meta-paths in heterogeneous information networks for representation learning, in: *Proceedings of the 26th ACM Conference on Information and Knowledge Management*, 2017, pp. 1797–1806.
- [42] Y. Cen, X. Zou, J. Zhang, et al., Representation learning for attributed multiplex heterogeneous network, in: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019, pp. 1358–1368.
- [43] B. Hu, Y. Fang, C. Shi, Adversarial learning on heterogeneous information networks, in: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019, pp. 120–129.
- [44] P. Wang, K. Agarwal, C. Ham, et al., Self-supervised learning of contextual embeddings for link prediction in heterogeneous networks, in: *Proceedings of the 30th Web Conference*, 2021, pp. 2946–2957.
- [45] Y. Lu, M. Gao, H. Liu, et al., Neighborhood overlap-aware heterogeneous hypergraph neural network for link prediction, *Pattern Recognit.* 144 (2023) 109818.
- [46] H. Fan, F. Zhang, Y. Wei, et al., Heterogeneous hypergraph variational autoencoder for link prediction, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (8) (2021) 4125–4138.
- [47] D. Kingma, J. Ba, Adam: A method for stochastic optimization, *The 3rd International Conference on Learning Representations* (2015).



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